

Guide for installing, configuring and using Proton Mail with Postfix on Linux.

Send email from mailx, sendmail and system utilities using Proton Mail

This guide describes how setup up Postfix to work with Proton secure SMTP providing system, security and application administrators to use system utilities that are mail enabled and software like sendmail, postfix, mailx and mutt to send reports, alerts and other material to people that presumably care.

The SMTP service can only send mail using a custom domain registered in a Proton Mail account. If the requirement includes receiving mails use mutt and the Proton Mail Bridge.

There are other Linux command line mail services that can use SMTP credentials but they may have the limitation of not being able to be used with some Linux system utilities.

Summary

- ◆ How to test a SMTP TLS connection.
- ◆ Getting the Proton Mail SMTP token.
- ◆ Installing and configuring postfix, the mail.cf file.
- ◆ Enabling SASL security, a TLS requirement.
- ◆ Enabling rewriting of local mails to the SMTP token user, this permits system and application accounts to use the SMTP mail service, useful for automation.
- ◆ Section **Notes & Mail Management** covers:
 - The idiosyncrasies of using TLS/SASL enabled postfix;
 - How setting up a SMTP mail-out service can cause catastrophic failure of a Linux box;
 - Examples of Linux flavours having differing interpretations of what their command line mail facilities can do and some have replaced the old with utilities that are shinier.
 - Plus, other interesting notes.
- ◆ Where to locate info for basic trouble shooting and security forensics.
- ◆ An example using of using Ubuntu sendmail and mailx with a heredoc. Heredocs demonstrate the versatility of the email clients. There is plenty on the internet how use Linux mail tools.

Specific software installation instructions are not included in this guide because software package management varies between Linux flavours.

Test SMTP TLS connection

To test the SMTP TLS connection the following should return something like this if everything is OK.

```
$ echo "q" | openssl s_client -connect smtp.protonmail.ch:587 -starttls smtp
CONNECTED(00000003)
depth=2 C = US, O = Internet Security Research Group, CN = ISRG Root X1
verify return:1
depth=1 C = US, O = Let's Encrypt, CN = R12
verify return:1
depth=0 CN = protonmail.com
verify return:1
---
Certificate chain
 0 s:CN = protonmail.com
  i:C = US, O = Let's Encrypt, CN = R12
--- < plus a lot more >----
```

Proton Mail SMTP Token

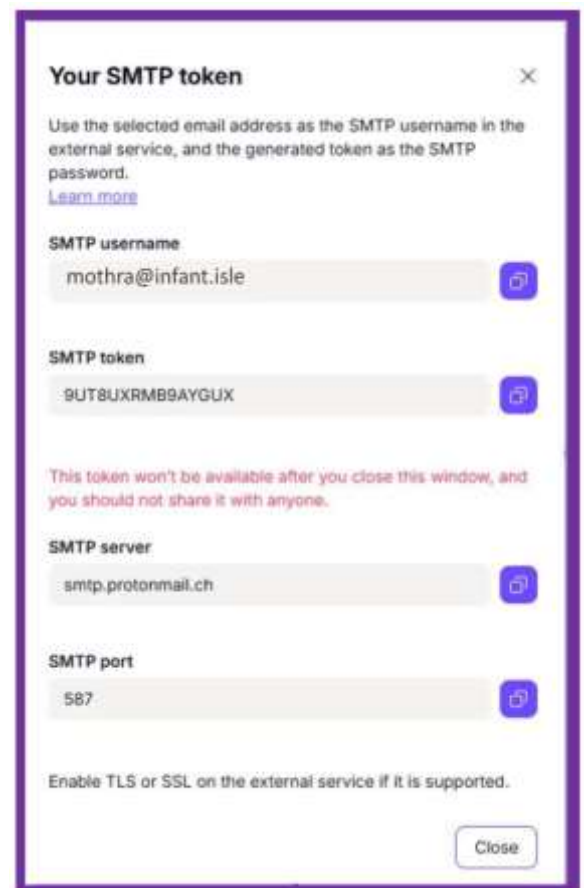
Obtain a Proton Mail SMTP token.

All Settings > IMAP/SMTP > SMTP Submission > Generate token

Copy its contents, take an image copy because when it is closed it cannot be retrieved. To recreate a SMTP token is trivial.

Proton Mail expects all mail incoming from postfix using this token to be from the SMTP username. If it is not from this SMTP username then it is rejected.

Proton Mail account **mothra@pm.me** has a custom domain name attached to the account, **infant.isle**



If a custom domain has multiple Proton addresses configured for it, every address can have its own individual token.

To send SMTP mails from the address *NoReply@infant.isle* requires that as a Proton Mail address, plus its SMTP token.

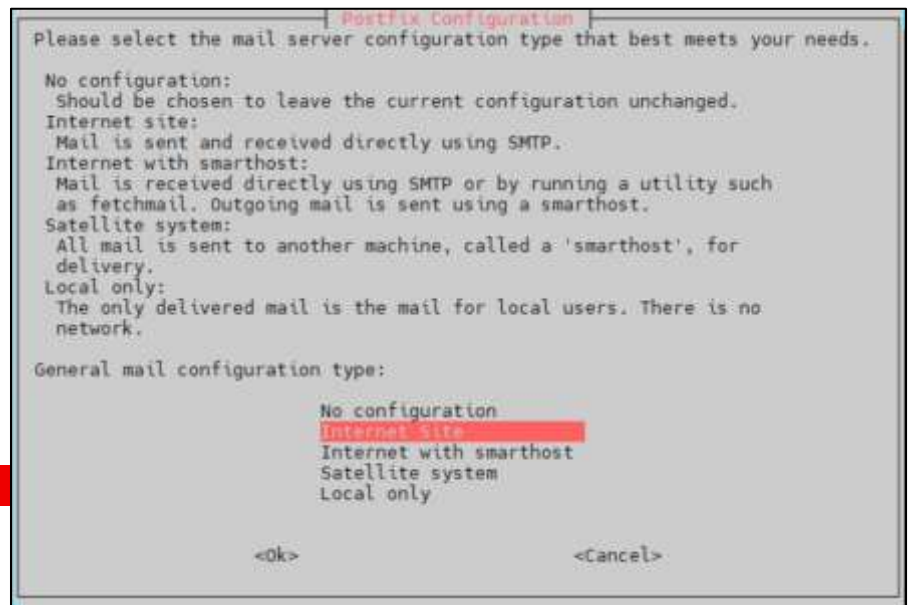
These tokens are not affected by changing the mail account password and can be terminated by **All Settings > IMAP/SMTP > SMTP Submission > Actions > Delete**

Postfix installation and configuration

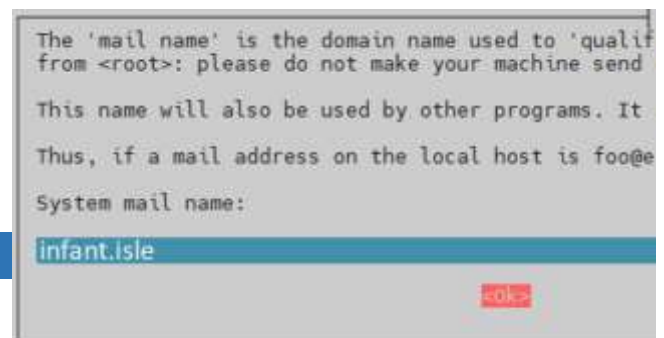
As root, install the **mailutils** package. This include sendmail, mail, mailx and other mail utliities. Ubuntu bundles postfix with mailutils. If postfix is not bundled with mailutils install that after mailutils.

The postfix installation requires the user to enter their System mail name with is the custom domain name.

Set mail configuration type to
Internet Site



System mail name to the
custom domain name



Once installation is complete the only directories of interest are

- Postfix configuration **/etc/postfix/**
- Postfix security configuration **/etc/postfix/sasl/**
- Postfix and mail logs log **/var/log/**
- All local mail accounts **/var/mail**

As the root user continue and go to **the /etc/postfix** directory to configure postfix

Make a backup copy of the original **/etc/postfix/main.cf** file before any changes

Edit the main.cf

The custom domain name(infant.isle) and hostname (fred01) have been added automatically during the installation process.

Remove all the comment lines, if required they can be referenced in the original main.cf backup.
Only add or change the highlighted lines.

```
compatibility_level = 3.9
myorigin = /etc/mailname
smtpd_banner = $myhostname ESMTP $mail_name (Ubuntu)
inet_protocols = all
mynetworks_style = host
mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128
mydestination = $myhostname, infant.isle, fred01, localhost.localdomain,
localhost
mailbox_size_limit = 0
alias_maps = hash:/etc/aliases
alias_database = hash:/etc/aliases
biff = no
recipient_delimiter = +
relayhost = [smtp.protonmail.ch]:587
cyrus_sasl_config_path = /etc/postfix/sasl
smtpd_tls_key_file = /etc/ssl/private/ssl-cert-snakeoil.key
smtpd_tls_cert_file = /etc/ssl/certs/ssl-cert-snakeoil.pem
smtpd_tls_security_level = may
smtp_tls_CAfile = /etc/ssl/certs/ca-certificates.crt
smtp_tls_security_level = encrypt
smtp_tls_session_cache_database = btree:${data_directory}/smtp_scache
smtpd_relay_restrictions = permit_mynetworks permit_sasl_authenticated
defer_unauth_destination
myhostname = fred01
inet_interfaces = loopback-only
sender_canonical_maps = hash:/etc/postfix/sender_canonical
smtp_sasl_auth_enable = yes
smtp_sasl_security_options = noanonymous
smtp_sasl_password_maps = hash:/etc/postfix/sasl/sasl_passwd
maillog_file = /var/log/postfix.log
```

The following have to added or edited in the /etc/postfix/mail.cf file.

Configuration item	Initial Setting	Required Setting
Relayhost	<i>Not set</i>	[smtp.protonmail.ch]:587
smtp_tls_security_level	may	Encrypt
inet_interfaces	all	loopback-only
smtp_sasl_auth_enable	<i>New</i>	Yes
smtp_sasl_security_options	<i>New</i>	Noanonymous
smtp_sasl_password_maps	<i>New</i>	hash:/etc/postfix/sasl/sasl_passwd
sender_canonical_maps	<i>New</i>	hash:/etc/postfix/sender_canonical
maillog_file	<i>New</i>	/var/log/postfix.log

Configuration item	Purpose
Relayhost	The destination for postfix mails. The same for all Proton SMTP tokens
smtp_tls_security_level	Encrypt outgoing mail
inet_interfaces	<i>loopback-only</i> blocks the receiving of external mail.
smtp_sasl_auth_enable	Configuration for the use of SASL authentication.
smtp_sasl_security_options	
smtp_sasl_password_maps	
sender_canonical_maps	This setting permits root, applications and local accounts to send mail.
maillog_file	Trouble shooting and security forensics.

Maillog_file

- The log file should only be accessible to root in a shared environment.
- Disable when not required, will require a postfix restart.

Enabling SASL security

Continuing as the root user, go to **/etc/postfix/sasl** directory, it will be empty. Create a **sasl_passwd** file with the following contents

```
[SMTP Server]:<SMTP Port> <SMTP User name>:<SMTP Token>
```

Using contents to the Proton SMTP token the result will be something similar to this.

```
[smtp.protonmail.ch]:587 mothra@infant.isle:9UT8UXRMB9AYGGUX
```

Run postmap command to setup the SMTP user account and password for postfix

```
postmap /etc/postfix/sasl/sasl_passwd
```

This will create **sasl_passwd.db** file in the same directory as the **sasl_passwd** file. By default, these files can be read by everybody, change them to root only readable.

```
systemctl restart postfix # RESTART Postfix
```

Enabling rewriting of local account mails to the SMTP token address

Security of the SMTP server requires that all mail must come from a single known source, the SMTP user on the Proton token. System users, application users and general users can be pre-configured in Postfix to use the Proton SMTP user name.

Create a **/etc/postfix/sender_canonical** file. The format of the file is two columns, the first is the local user name and the second is name that postfix will rewrite it to, the SMTP user name.

```
# Format of each line
# local_user SMTP_user
root mothra@infant.isle
morusra mothra@infant.isle
kyoga mothra@infant.isle
```

Every time this file is edited the map has to be compiled and reloaded into Postfix with the command

```
postmap /etc/postfix/sender_canonical
```

This will create **sender_canonical.db** in the **/etc/postfix** directory.

```
systemctl restart postfix # RESTART Postfix
```

All done, ready for testing.

Notes and mail management

- Remove the STMP user name from the Proton Mail Contact List. Once removed the Display Name will be listed as the Linux user account that sent the mail. If present in the Contact List the mails will have no Display Name.
- Cc and Bcc Do not work as expected in postfix, the default settings of postfix treat all mail addresses the same. If a mail from sendmail or mailx is feed into postfix any addresses assigned to **Cc** and **Bcc** will be processed as **To** addresses. Using **sender_bcc_maps** to enable Bcc has a scope of usage too narrow to make it useful.
- Check your Linux flavour for mail utility options, they vary. Ubuntu mailx does not have **-c** (Cc) and **-b** (Bcc) options and **-A** is used for attachments others use **-a**. Suse Linux mailx has the **-t** option it is absent in Ubuntu. Red Hat has replaced mailx with s-nail.
- From (-r) mail addressing in the formats will be sent "No Reply <mothra@pm.me>" and "<mothra@pm.me> (No reply)". The "No Reply" will be discarded, the recipient only sees *mothra@pm.me* with the display name being the Linux user account name. Best to go with -r mothra@pm.me format with no decorations.
- If the sending user has been configured in **sender_canonical** there is no requirement for the -r option to be used. The -r option is required for all uses that are not in the /etc/postfix/sender_canonical file. The -r option may be substituted with the -f option, -F cannot be used – See **Example – Sending Mail**
- The **cron** task scheduler will send STDOUT and STDERR text to the users mailbox if that output is not directed to an app/program, file or the /dev/null back hole. The app can be a mail utility like mailx or sendmail which engage postfix for delivery.

If the cron user is setup in **sender_canonical_maps** the environment variable MAILTO can be used to send system mails directly.

```
# Cron resets MAILTO environment variable when it scans from top to bottom.
MAILTO=sysadmins@overthere.com
0 3 * * 0 /usr/local/bin/weekly-reports
# Reset MAILTO somebody else, the backup crew
MAILTO="bkups@overthere.com"
0 20 * * * /usr/local/bin/bkups
# Disable email output for the rest
MAILTO=""
* 1 * * * /usr/local/bin/whatever > /dev/null 2>&1
```

If your script does not work in cron, run the *env* command, most of that output will be missing for cron. Cron runs in a minimal environment, consult an internet search for more information,

Local Linux mail requires management.

Automated mailing and manual command line mailing requires monitoring and management of logs plus the mail queues and the mailboxes in the /var/mail directory.

Non-delivery reports and other error mails accumulate in user mailboxes, which could cause catastrophic failure of applications and server functionality if a filesystem filled up.

The *mail* utility can run interactively to remove mails in bulk (d *). It is safe to delete root and any other uses mail file in the /var/mail/ directory. A new file will be created when a mail arrives for that user.

Mails with invalid options that are discarded by postfix may not appear in the Linux user's mail box. If postfix logs are enabled, check them for "rejected" mails.

Basic troubleshooting and security forensics

Postfix log /var/log/postfix.log has better detail has better information than the /var/log/mail.log, additional information can be found in the systemd logs

```
grep pickup /var/log/postfix.log | grep -v 'SMTPuser' # see who has been sending mails
journalctl -f -t postfix/smtpd # smtp specific info
```

Example - sending mail

Internet examples on how to use mail, sendmail, mailx and cron's MAILTO facility are common these are the basics for using a heredoc to send mail.

```
#!/usr/bin/bash
#sudo /usr/bin/bash -c 'echo "===== $(date) =====" \
>> /var/log/postfix.log' 1

function sys_storage(){
# Filesystem capacity use
echo "-----"
df -h | awk '{printf "%7s %7s %7s %7s %s %s\n", $2, $3, $4, $5, $6, $7}'
echo "-----"
df --total -h | grep total | awk '{printf "%7s %7s %7s %7s %s\n", $2, $3, $4, $5, "TOTAL"}'
} #-----
recpt_addr="godzilla@bass.isle"
cc_addr="gigan@xilien.co" 2
bcc_addr=""

/usr/sbin/sendmail -i -r mothra@infant.isle -t <<- msg_end
To: $rec_addr 3 4 5
Cc: $cc_addr
Bcc: $bcc_addr
Subject: "$(uname)local mail capacity report $(date)" 6

Number of mails in the queue is .. $(mailq)
Number of mails in $USER mailbox . $(grep -c "^From " /var/mail/$USER)
Size of $USER mailbox ..... $(du -h --apparent-size /var/mail/$USER | awk '{print $1}')
Total size all mailboxes ..... $(du -h --apparent-size /var/mail/ | awk '{print $1}')

Filesystem Storage
$(sys_storage)

Produced by $USER 6
msg_end
```

1. During testing, time stamp the postfix log, this makes the test obvious. Logging must be enabled
2. The addresses in **Cc:** and **Bcc:** will be processed as **To:** addresses. Postfix doesn't discriminate; they are all the same.
3. The option **-i** suppresses of the "dot on a line by itself" problem. A solitary dot on a line will terminate body input and initiate the sending of the mail, problematic if a command or application produces that as an output line.
4. The option of **-r <SMTP user>** is always required if there is no entry for the user in the **/etc/postfix/sender_canonical** file.
5. **-t** is a requirement for sendmail to accept To, Cc and Bcc from STDIN.
<<- starts the heredoc, **"-"** (optional) removes unwanted indentation, **msg_end** is a tag of your choice.
6. It is helpful to include information about the source of the mail in the subject or body.
7. In Ubuntu mailx will be almost identical to this, **-t** not required, if present its ignored.